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Codiscoverer of uranium-233. He has written several books and published numerous papers on the subject of nuclear power.

I'm just a labaholic. I always liked the idea of working in a laboratory. If somebody locked me up in a lab and just sent food in, I'd stay there forever.

Even though I was doing well at Western Reserve Medical School—this was during the depression—I felt the need for more chemistry. So I took a leave of absence from med school in Cleveland and went to the University of California in Berkeley. It was in physical chemistry, my major interest. The giants in the field were here, and I was scared to death. I felt I didn't know anything.

A new graduate student would go around and ask any professor if there was a problem he could work on. I chose Glen Seaborg,* who was working on artificial radioactive activity. I thought this would be useful in medicine when I got back to Western Reserve.

Seaborg assigned me to find out if a substance known as protoactinium-233 existed. I proved that it did exist and how to measure it. We published a paper in the journal *Physical Review*. Then I figured since it emitted the radiation we called beta particles, it must be decaying into uranium-233. I started to search for any decay of uranium-233, because we didn't know if it would last seconds or minutes or thousands of years. It turned out it lasts 125,000 years. This was 1940.

The teaching assistantship paid \$65 a month. I had just married, so after paying the rent, \$35, we had \$30 to live on. There was no job for the summer. Seaborg tried to raise \$100, so I could stay and work on the problem. He failed. Our discovery of uranium-233 was, afterward, hailed as a development worth fifty quadrillion dollars, if nuclear energy went ahead.

We bummed a ride back to Cleveland and stayed with our parents until the fall, when my teaching assistantship resumed. In that short time, things had changed. Scientists were considering work on the possibilities of a bomb, because, after fission was discovered in 1939, there was concern the Germans would beat us to it. This was '40, '41. I'm back in Berkeley and we got some money.

"In nature, there are three kinds of uranium. U-38 constitutes about 99.3 percent of the uranium you dig out of the ground. It will

* Winner of the 1951 Nobel Prize in chemistry.

not undergo fission with slow-moving neutrons. U-235 would split apart with slow neutrons. This fission is what makes our nuclear reactors and bombs possible—the one dropped on Hiroshima, for example. What makes our discovery a key is that element 90, thorium, with neutrons added, creates uranium-233. So not only is the world's uranium supply available for nuclear power or bombs, so is thorium's. Furthermore, with uranium-233, fission can occur with slow or fast neutrons. That's how come it was regarded as a fifty-quadrillion-dollar discovery.

“Enrico Fermi, at the University of Chicago, proved with his reactor that plutonium, element 94, could be produced. The nice thing about plutonium is that it's easier to separate it from uranium than to separate uranium isotopes from each other.

“In those days, there was not enough plutonium to see. We could only track it by its radioactivity. So we set aside our uranium-233 and I became the fourth chemist in the world to work with plutonium.”

The Manhattan Project began in '41, '42. Before Oppenheimer went to Los Alamos, he expressed interest in my measurement data on the fissioning of uranium and plutonium. When General Groves became the head of the project, everything changed overnight. Before, when we wanted something, we had to wait six months, because the war was on. The moment the Manhattan Project came into being, we had Triple-A priority. We could take things right off the trains.

The Hanford, Washington, site was chosen for reactor work in isolation. Oak Ridge, Tennessee, was chosen as the intermediary site for uranium separation. Oppenheimer chose Los Alamos, New Mexico, for the fabrication of the bomb itself. He invited all of us in the Berkeley chemistry group to go with him.

We were very dedicated workers in those days. Oppie said, “We won't guarantee you'll have any contact with the outside world for the duration of the war.” Because my wife was in medical school at the time, I elected to stay in Berkeley.

Two months later, Oppenheimer came to see me in Berkeley. He brought Joe Kennedy with him. Joe, head chemist at the Los Alamos lab, was the most brilliant chemist I ever met. You'd go out to radio shops, get him parts, he'd put it together and make it work. His hands were golden. He died at age thirty-eight, of cancer of the stomach. It may have been from radiation, we can't be sure.

Oppenheimer said, “We need half a milligram of plutonium.” The world's supply at that point was one-twentieth of a milligram. Today, we're worried about where we'll put fifty tons of plutonium. I said, “You'll have lots of it from Oak Ridge in about nine months.” He said,

"We need it now for some special measurements. It will make a big difference on whether or not we can make the bomb." Joe Kennedy had told him that I'd worked out a way of separating plutonium from uranium, and from all the garbage that gets produced. I had worked something out with little test tubes and beakers on a lab bench.

If we were going to make a half milligram of plutonium before the reactors produced it, we'd have to use the Berkeley cyclotron. We'd have to bombard a ton of uranium, running the cyclotron for six, seven weeks. Oppie said he had already worked it out with Ernest Lawrence,* the new head of the Berkeley labs. Seaborg had gone off to work on plutonium in Chicago.

We ran the cyclotron for seven weeks, day and night. Then we let it cool off, because there's a lot of short-lived radioactivity after such a bombardment, very dangerous. But since time was of the essence, we let it cool for only a few weeks instead of months. Working around the clock, we ended up—after using a ton of uranium—with about a quarter of a teaspoon of liquid plutonium. It turned out we had 1.2 milligrams. Oppie was elated. "You can keep two-tenths of a milligram to play around with. We'll take the one milligram." They took it down to Los Alamos and got the measurements they wanted. It seems I'm now the fair-haired boy in the atomic energy arena.

In 1942, I was twenty-four. I took the war very seriously. I wasn't questioning anything. I *did* have one question. I asked Arthur Wahl, a fine chemist who shared the lab with me, "Suppose the bomb is made. How will people live in the area with all that radioactivity around?" He said, "If a bomb goes off, it's going to send this material up into the stratosphere and is never coming back." The idea of fall-out hadn't occurred yet. It turned out later that *all* of it came back.

For me, the excitement had already worn off. I went back to med school—the University of California Medical School in San Francisco.† During the war years, I had established a lot of capital for the work I'd done. I had all sorts of offers from the Mayo Clinic. Oregon, Washington University in St. Louis, others. I accepted an assistant professorship at Berkeley in Medial Physics, a new field.

I wanted to do something with either cancer or heart disease. I chose to work with large molecules because I had had some ideas about cholesterol. This was 1948. The terms "good cholesterol" and "bad cholesterol" came out of our work. I'm still equivocal about good cholesterol, but there's no doubt about the bad being bad.‡

* Winner of the 1939 Nobel Prize in physics, for his invention of the cyclotron.

† He won the Gold-headed Cane Award as the student who most exhibited the attributes of a true physician.

‡ The American Heart Association awarded him the Lyman Duff Lectureship, their highest award. "I'm now, it seems, a fair-haired boy in heart-disease work."

Just last year, in 1993, the Heart Association Arterial Sclerosis Council chose me as honored guest speaker. I said I didn't want to simply engage in recollections. I had a lot of data I hadn't yet analyzed. I worked my butt off for three months and represented it at the meeting. It was good to get back in that field a little.

By 1954, I was a full professor of medical physics at Berkeley. What happened? I hadn't done much radiation work at all, but that year Ernest Lawrence and Edward Teller had established the Livermore branch of the Berkeley labs. In 1952, they decided to speed up work on the hydrogen bomb. Teller was unhappy with the rate at which Los Alamos was moving. He wanted it faster.

At about that time, I lectured to a group of physicists about my work in heart disease. Lawrence was in the audience. I went over it lightly. When I finished, Lawrence stood up and said, "I don't get it." The others hadn't said a word. About ten days later, he pokes his head in the door of my lab and asks me to explain my lecture further: "Tell me more." Finally, with the enthusiasm of a little kid, he shouts, "I got it!"

He asks me if there is anything I need in furthering my heart-disease work. I need an ultracentrifuge; it separates proteins from molecules. It costs about \$16,000. Today it would cost \$150,000. The business manager of his Berkeley lab says it's not in the budget. Lawrence said, "I didn't ask you about the budget. Get John an ultracentrifuge. That's an order." That's the kind of guy he was. From that time on, he'd bring prominent guests to see our work. He was so enthusiastic, he'd take over and tell them all about it.

One day, in late '51, he calls me into his office. "I'm worried about those guys at Livermore. They're so intense in work on the hydrogen bomb. I don't like dangerous things done in the lab." He asks me to visit Livermore, "roam around and if you don't like anything you see, tell them to shape up, because you represent me. I'd feel a lot more comfortable about their health out there." So I organized the Industrial Medical Department at Livermore.

What happened during those years was crucial. Their major work was in weapons research. I helped them with some of their considerations for the Pacific tests, but I still wasn't doing anything on radiation. I was arranging the examinations of the lab guys who were coming back from the Marshall Islands, but I was not yet questioning. In 1957, after three years at Livermore, I came back to Berkeley.

I was very stupid in those days. In 1955, '56, people like Linus Pauling were saying that the bomb fallout would cause all this trouble. I thought, "We're not sure. If you're not sure, don't stand in the way of progress." I could not have thought anything more stupid in my life.

The big moment in my life happened while I was giving a health lecture to nuclear engineers. In the middle of my talk, it hit me! What the hell am I saying? If you don't know whether low doses are safe or not, going ahead is exactly wrong. At that moment I changed my position entirely.

I was going ahead with my heart-disease work. Comes 1962. The Russians break the agreement between Khrushchev and Eisenhower on the moratorium on atmospheric testing. JFK, now in office, tells the labs to go ahead, to show the Russians we'll not be second best in nuclear weapons. The Livermore lab goes gung-ho. Many nuclear bombs, different sizes, different types, are tested in the atmosphere. They put on a big show in the Pacific, in the Nevada test site, which causes fallout all over the United States.

I get a call from John Foster, head of the Livermore lab. It's Teller, behind the scenes, who calls the shots. Foster says, "We have a strange request." The Atomic Energy Commission would like to set up a biological medical unit here. I said that we have eighteen, nineteen laboratories already. Why another one? He explained: from the Nevada testing site, we had released so much radioactivity in the state of Utah that the milk was three times the safe level. The problem was "solved" by the Federal Radiation Council, the group that was supposed to set the standards. They suddenly discovered that three times as much as what had been previously considered safe was still safe. They raised the tolerance level, it was a fraud.

I told him, "If you're gonna explode bombs in the atmosphere and the winds blow, the materials are gonna go deposit somewhere. I don't see how having biologists and medical people in the lab is going to solve the problem. This is just to help the Atomic Energy Commission get off the hot seat."

He agreed that fallout was the problem. He promised me a new building with great facilities and that I could bring in any scientists I wanted.

"I don't trust the AEC," I said.

"I don't either," he said. "If you see anything you don't like, we won't let the AEC stifle you. We'll back you all the way." My mission was to evaluate all the aspects of bomb testing.

I insisted on a letter from Clark Kerr, then president of the University of California, that if I didn't like anything here, I could have my full professorship back. The letter was written. The university regents agreed, as well.

I went to Washington to sign the papers. Glen Seaborg, my former mentor, was now the chairman of the Atomic Energy Commission. Kennedy had appointed him in '61. I told the commissioners that I wasn't sure about accepting this job. "If any of your programs

are bad for public health, I will publicly say so. I will not tolerate any interference by documents stamped 'secret.' If you have any doubts, get someone else for the job." Seaborg said, "Jack, all we want is the truth." In retrospect, if there was one thing they didn't want, it was the truth.

They built us a brand new building. I had the loveliest office and special laboratories for myself. I had 125 people. I was working, happy, teaching a little. Some of the work on cancer and chromosomes was getting exciting. I asked to be excused as the associate director so I could work at the lab more.

There was another program at Livermore: Plowshares. It was Teller's favorite project: to move mountains, diagram canals, and divert rivers with nuclear explosives. They were to go to Panama or Colombia and build new canals with hydrogen bombs. They planned to use 315 megaton bombs in a row to excavate the land. Each bomb had one hundred times the explosive force of the one that fell on Hiroshima. When Foster told me about it, I said, "Are you serious?" He said, "It's Teller's baby. We want you to evaluate its health effects." I put people to work on it, but at a director's meeting, I called it biological insanity. It did not endear me to Teller.

Plowshares was canceled, not for reasons of health. The government was promoting the nonproliferation pact, and this project could be misinterpreted. In any event, my open criticism of it made Teller less enamored of me than ever. It was about this time that rumors began about me as the enemy within.

I further aggravated some people in 1969. I was invited to give a lecture for nuclear engineers on any subject I chose. My talk was "Low-dose Radiation, Cancer, and Chromosomes." My calculation was this: If everybody in the country were to get the radiation dose that was permissible under the existing standards, there would be sixteen thousand extra fatal cancers per year. I offered one further caveat: If there is an official safe dose of radiation, we'd have to reduce that number, because we must assume there is no safe dose. The moon shots were going off at the time, so the speech got no notice.

About a week later, the new head of the Livermore lab, Michael May, said, "Jack, I've got some flak from Washington. They're upset with the talk you gave." I said, "I didn't give away any secrets. I gave a calculation." "They're upset because they didn't know about the talk in advance. They were blindsided by the reporters." Would I mind, in future talks, sending a copy in advance. I said sure, I had no reason to withhold it.

I had an invitation from the American Association for the Advancement of Science to take part in a symposium on nuclear

power. I suggested Arthur Tamplin, head of one of my divisions, who was quite familiar with it. He gave an advance copy of the speech to May.

Three days later, Tamplin comes in mad as hell. He threw the manuscript on my desk. "Look at this." The entire guts of the talk was penciled out. They left in the prepositions and the conjunctions. May had told him that if he persisted in giving the talk as outlined, he couldn't identify himself as a member of the Livermore lab, he couldn't use lab secretaries to type his manuscripts, and would have to pay his own travel expenses.

I told May this was in violation of everything agreed to when I came to Livermore. I was going to call the guy at the Triple-A 5 and tell him that he was censoring Tamplin and that I now consider Livermore to be a scientific whorehouse. I did call the association guy, who was scared to death. May stormed out and didn't talk to me for months. He backed off. Tamplin made a few minor changes and his speech went off as scheduled: critical of nuclear power.

Meanwhile, I had accepted an invitation from the Muskie Committee on Atomic Energy matters. At this time, I testified that our calculation showed that it might be closer to thirty-two thousand extra deaths a year rather than sixteen thousand.

Now the AEC went into high gear to get us. "Gofman's an incompetent," they told the reporters right after the Muskie testimony. For the last seven years, they had given me three and a half million a year and a new building and, suddenly, I was incompetent.

Chet Holifield, a congressman from California, was the new chairman of the Joint Committee on Atomic Energy. He sent word that he wanted to see Tamplin and me. He led off, "Just what the hell do you two think you're doing?"

"What do you mean?"

"You're getting all the little old ladies in tennis shoes up in arms about our program. Why the hell are you doing this?"

One of his aides whispered, "These are two highly respected scientists."

"I don't give a good goddamn who they are. They're interfering with our nuclear energy program."

I said, "Mr. Holifield. Our job is to find the health effects of radiation. We've worked on it, and this is our information." He said, "Listen, I've been told that if we give people a hundred times the amount of the radiation dose we allow, nobody will be hurt."

"Who told you that?"

"The Atomic Energy Commission, that's who. Guys like you have tried to interfere with our program before. We got them and we'll get you."

Those were his parting words.

It turned out that their expert authority was a physicist who had been dealing with data of the early '20s on people painting luminous radium dials. There were women in factories who were getting horrible bone cancer. He was saying that at a thousand rads nobody would be hurt. It was absurd, a horrible misanalysis of data. He told this to the AEC and they *loved* it. It was the answer they wanted. They didn't give a damn whether it was true or not.

From there on out, things got rougher and rougher. In early '70, I wrote Seaborg a letter. I remember him as the man who shared credit with me, his student, as codiscoverer of uranium-233. I thought he'd still have some respect for me. I warned him of some of his people and the dangers to public health and of his promise to seek out the truth.

He didn't back us. He just melted away. He didn't say anything nasty about us. He didn't have to. His underlings did the job. The atomic electricity industry was not too friendly either. One of their guys told a reporter. "Gofman's just been released from a mental institution." She said, "Oh, that's very interesting. I'll write it up in the *Village Voice*." [Laughs.] The guy spent the next half hour pleading with her not to put it in. The press was generally fair to Tamplin and me.

The battle was just heating up. I was still working at Livermore, had twelve people on cancer, chromosomes, and radiation. One day, the associate director told me that the AEC ordered them to take away my research money: \$250,000. The lab, at first, refused. They said my work was first class. A year later, the AEC renewed its demand. I didn't fight it. I was thinking about those twelve people who would lose their jobs if I insisted on keeping my program going. It was a big mistake. I should have raised holy hell and taken it up with the university at the highest level. I said, "Let's kill the program." We did that day. It was a good program and the public would have benefited. We had already published some things in the *British Cancer Journal*. This was 1972.

I thought this would be of real interest to the National Cancer Institute at Bethesda. I was pretty sure they'd come up with \$250,000 a year. The director sounded enthusiastic. He said, "Your program is exactly what we need. Would you be willing to cooperate with the Yale people on this work?" I said I'd love to. We talked for several hours. He'd let me know in three weeks. It looked very good.

About eight weeks later. I sent a nice note off to the director, understanding that there might be the usual delays in this matter. But I had to decide pretty soon about moving all our gear out of Livermore. I got a letter back from a third-echelon deputy. "Many

thanks for your inquiry. The work you are proposing is not in the main line of our interest, but if you have any other ideas, please let us know." I never heard from the director.

I can guess what happened. I think people within his shop said, "Look, this character Gofman is giving the Atomic Energy Commission fits. What the hell do we need him for? He's trouble."

I applied for grants everywhere, the National Heart Association among them. I got nowhere. I had a good name in the field, so what? I no longer had any funds for research. So I resumed teaching again at Berkeley in the Division of Medical Physics.

After a couple of years, I decided to take an early retirement. I'm now a professor emeritus of molecular and cell biology. I serve on university committees. I give some lectures. The Department of Medical Physics no longer exists.

I became active in the anti-nuclear power movement. The power people were constantly putting me down because my work was harmful to them. If they couldn't sell the idea of safe doses, they weren't going to last forever. I talked at rallies and meetings all over the country and abroad. I wrote a book called *Poisoned Power*, which became sort of a bible for the antinuclear movement. I published a little collection of talks, *Irrevy: An Irreverent Illustrated View of Nuclear Power*.

There is one organization that may be worse than the AEC. It's the Veteran's Administration. There are so many cases of all these servicemen who've been out in the Pacific and flown through all these atomic clouds. They developed leukemia and cancers.

Orville Kelly had been on the Bikini Islands in connection with the Pacific tests. He developed lymphoma. He wrote me, "I've been through appeals to the VA and got nowhere." He had his records. I wrote a detailed report on him to the VA and they rejected it. [*Angrily.*] Orville Kelly was the first guy to get money. His wife benefited from it when he died.

Another woman, Shirley Donahue, wrote me. She had two young children. Her husband had been a lieutenant colonel in the air force. They used to fly right through the clouds to collect samples of the air. These guys got a hell of a dose. Shirley's husband had an accumulated dose of seventy-five rads, whole body radiation. He developed leukemia. My calculations show that it was due to his experience on the job and nothing else. His case was refused. We went through it three times. He was dead. Shirley is finally getting a pension from the VA.*

* "As you enter John Smitherman's house in Mulberry, Tennessee, he calls out to you. You immediately see a large, handsome face. Simultaneously, you see that he is legless. It is his left hand, resting on the arm of his automated wheelchair,

The next thing was the Karen Silkwood case. They needed someone to testify at the trial on the hazards of plutonium. I had written an article for the *Journal of the American Medical Association*. Silkwood, you remember, was killed in an auto accident. "Will you help serve as a witness?" I met with Gerry Spence, the Silkwood lawyer, the night before I was to testify. He said, "Watch the jury. They'll tell you when you've made your point."

Kerr-McGee, the company being sued, had a bevy of lawyers, about a dozen pin-stripes. One of their lawyers asked, "Could you explain how it is we get such different answers from illustrious committees?" He goes on to name them. He made the mistake of giving me an open-ended opportunity to talk. I talked about an hour, and the guy was speechless. He couldn't stop me because he had asked the question. The jury was lapping it up. We won the case. Her family was awarded ten and a half million dollars. The Supreme Court sent it back, and they finally settled for \$1.4 million.

I entered a case on behalf of the downwinders in Utah. Most of the winds were blowing in that direction from the Nevada tests. The government was saying, "No problem. It's not going to hurt you." Cases of cancer, leukemia, started coming up. They agreed to have twenty-four people represent the twelve hundred who claimed injury. I testified in that case. We beat them. The government was furious, especially when the judge cited my book as one of the three definitive works on the subject: *Radiation and Human Health*. They *hated* it. We won the case, but it was reversed by the Supreme Court on the grounds of discretionary function. If somebody like the

that draws your reluctant attention. It is at least five times normal size; roughly ridged, corrugated, grayish; it resembles an elephant's trunk.

"There was 42,000 veterans that participated in Operation Crossroads; 27,000 of 'em is dead. That was in the Marshall Islands and the Bikini atoll. We were advised there would be nothin' harmful. Just a lot of excitement and a lot of fun.

"We were circlin' around the USS *Mount McKinley* when the bomb exploded. We were standin' in shorts. I had a T-shirt on, just like this. The brass stayed under cover the whole time. They wore heavy clothing. They were protected all the time.... All of a sudden, we saw this huge ball of fire come from the bottom and go up. We felt the heat, we felt the shock wave....

"Comin' back from Washington, D.C., I tried to go to the VA hospital to see a doctor. They refused to admit me because I wasn't there on a scheduled appointment. At a private hospital, they found cancer of the colon and liver, terminal. I've been fighting with the VA to gain service connection out of this whole thing. They have turned me down for the sixth time.

"They admit I was exposed to radiation, but not enough to create a problem. Three doctors on my behalf said I was exposed anywhere from 1,000 to 1,800 rads."

—From *"The Good War": An Oral History of World War II*, 1984

John Smitherman died on September 11, 1983, a few months after my visit. His widow, Rose, had been denied any compensation by the Veteran's Administration for years. Recently, I have been informed, she is receiving something.

president decides a crisis exists and he doesn't go through the necessary things, say, to protect people, he could say, "I used my discretion for the good of the country." This could be extended to anybody in the government who was supposed to monitor the doses. You couldn't sue them because of their discretionary function. It meant you could get no relief from the government at all in the law courts.

That's when I quit. I decided to write a book on radiation and human health. Somebody has to do it and it might as well be me. If it isn't done we'll build up a bunch of lies in textbooks and have the whole thing whitewashed. So I said, "No more law suits, no more rallies, no more lectures, I'm going to work on research"—and I'm still doing it.

It's hard to work in science when you discover there's very little you can trust. There's a group called the Radiation Effects Research Foundation in Japan. It was once called the Atom Bomb Casualty Committee, to evaluate the effects on Hiroshima. It was cosponsored by the Japanese Ministry of Health and the U.S. Atomic Energy Commission. It never occurred to me to question them.

They came to the conclusion that there are fewer neutrons in the mixture of radiation than we thought. They were going to assign new doses to people. They can make data look like anything they want. I was appalled. They were shuffling the whole deck with nobody watching. It was the clearest-cut violation of the cardinal rules of medical research. I wrote to the commission's chairman: "Dr. Shigematsu, you cannot do this. It violates all the rules." They had pulled out fifteen thousand records. It was an open invitation to fudge the data. He replied, "I appreciate your concern. It's valid. Just trust us." The reason we have rules is that nobody has to say "trust us."

How does it happen that most of these reports came out favorable to the nuclear industry? If we allow things to go on unchallenged, here's what's going to happen. Ten million, a hundred million, or a billion people are going to suffer from cancer and genetic diseases because the lies will become the truth concerning radiation and health. How many colleagues have I convinced? One in a hundred, perhaps. If I had unlimited access to Treasury bills, I could have reversed it overnight. It all depends who has the money as to what scientists think. The people at the Livermore lab saw what happened to me. Do you think any of them speak out? The cards are stacked.

Radiation will be regarded as harmless. It may even be good for you, because we've proved it harmless. You don't have to worry about waste, because we can put it out in landfills. So for a hundred years, lies will be truth and truth will be lies. The cost will be the multimillions who will suffer from vicious diseases.

Last year, in Stockholm, I was given an award by the Right Livelihood Foundation. I shared it with a courageous Russian journalist, Alla Yaroshinskaya, who blew the whistle on the lies his government was telling about Chernobyl.

I proposed in that award talk that if you want the truth, 10 percent of the money that the government provides for health research should go to grass-roots citizens' groups, who will pick their own scientists. The ordinary person, the layman, needs to say, "We'll support the scientists who dedicate their lives trying to ascertain the truth."

They should have full access to all the data and their job is to blow the whistle on Chernobyl, Hiroshima, Three Mile Island, or whatever. If you don't do this, the price you will pay will be biomedical unknowledge. Truth will become falsehood, and falsehood, truth.

The nuclear power movement is worldwide. You couldn't have told the difference between capitalist and Communist countries. I'm talking about nuclear power, not just bombs. I'm still speaking against it, writing against it, and blowing the whistle.